Band-1 Hadamard Order 5812 — Miyamoto (Paley-I H_1452 + C_1454 over GF(1453)); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_1452/C_1454/H_5812 Miyamoto PASS; Paley-I/II SKIP; 5812 Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5716_Miyamoto`; `2026-09-15_Band1_Hadamard_5476_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5812_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1452 and 5812 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with preferred Paley-I seed H_{1452} = Paley-I($q = 1451$) and symmetric Paley conference C_{1454} over **GF(**1453**)** (prime field, $q = 1453 \equiv 1 \pmod 4$). Double path for H_{1452} is **SKIP** ($q = 725 = 5^2 \cdot 29$ not prime). Classical SPK paths **FAIL**: Paley-I ($q = 5811 = 3 \cdot 13 \cdot 149$ not prime); Paley-II ($q = 2905 = 5 \cdot 7 \cdot 83$ not prime); Kronecker (4×1453). Therefore order **5812 is a Miyamoto-only bonus (NOT in SPK)** (same honesty pattern as 5476). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`.

Abstract

Finite constructions of orders **1452** and **5812** via Miyamoto.

- H_{1452} = Paley-I ($q = 1451$) — preferred; Double SKIP ($q = 725$ not prime)
- C_{1454} = Paley conference over GF(1453) prime field
- H_{5812} = Miyamoto(H_{1452}, C_{1454})
- Paley-I / Paley-II for 5812: **SKIP**

SPK: 5812 Miyamoto-only bonus (NOT in SPK). Seed 1452 enters SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5812.py`

Results. overall_pass=true; units 67/67; wall 160.0103s; all max|HH −nI|=0.

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Arithmetic

- Target $n = 5812 = 4 \cdot 1453$; 1453 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1452} and C_{1454} (Paley conference over prime field 1453).
- Seed preferred: Paley-I $q = 1451$ (prime $\equiv 3 \pmod 4$).
- Double SKIP: half=726, $q = 725$ not prime.
- SPK: Paley-I $q = 5811$ **FAIL**; Paley-II $q = 2905$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.

Constructions (verified)

Object	Method	max|HH −nI|	Wall (s)
H_1452 Paley-I	Paley-I q=1451 (preferred)	0	2.551
H_1452 Double	SKIP q=725 not prime	—	—
C_1454	Paley conf GF(1453) q=1453	0	2.4389
H_5812 Miyamoto	Miyamoto(H_1452, C_1454)	0	152.4475
H_5812 Paley-I	SKIP q=5811 not prime	—	—
H_5812 Paley-II	SKIP q=2905 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **5716** + **1428** (from 5716 run) + verified **1452**; **5812 NOT folded** (Miyamoto-only); **exclude** Miyamoto-only bonuses including **5476/5284/5812**. Known count 1201. Next constructive gap **2052** (OPEN).

Quantity	Value
5812 in SPK	**False** (Miyamoto-only bonus)
1452 in SPK	**True** (Paley-I)
5716 in SPK	**True** (prior Paley-II)
5476 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P NP/YM/Frankl/NS/AC/Clay-closed claims.